

PATRICIA MAGUMISE

Data Analyst / Data Scientist

patriciamagumise26@gmail.com | linkedin.com/in/patriciamagumise

PROFILE

Data Analyst with 5+ years of professional experience in database administration, data governance, and records management, now completing an M.S. in Applied Data Science (4.0 GPA) at Clarkson University. Proficient in Python, R, SQL, PySpark, Tableau, and Power BI, with hands-on experience in statistical analysis, machine learning, data visualization, and database design. Proven ability to turn large, unstructured datasets into accurate, decision-ready information.

EDUCATION

M.S. Applied Data Science | *Clarkson University, Reh School of Business - Potsdam, NY Expected Spring 2027*

GPA: 4.0

- Coursework completed: Database Modeling & Design, Probability & Statistics for Analytics, Data Warehousing, Information Visualization, Data Mining, Machine Learning, Big Data Processing, Natural Language Processing, and Data Driven Application Development

M.S. Library and Information Science | *National University of Science and Technology - Zimbabwe 2022*

B.S. (Honours), Library and Information Science | *National University of Science and Technology - Zimbabwe 2019*

TECHNICAL SKILLS

- Data Analysis & Programming: SQL, Python, R, PySpark, Statistical Analysis
- Visualization & BI: Tableau, Power BI, Data Visualization
- Data Management & Governance: Database Design, ERD, Data Cleaning, Data Quality, Records Retention, Data Entry, Confidentiality & Discretion
- Other: Koha Library Management System, Microsoft Office
- Languages: English (fluent), Shona (native)

PROJECTS

LA Crime Prediction - *Clarkson University*

- Built and evaluated a gradient boosting model on a dataset of 733,000+ crime records to predict incident patterns, achieving 0.78 AUC.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | *Clarkson University - Potsdam, NY Sep 2025 – Present*

- Manage institutional databases of 100,000+ records, applying data validation and quality-control protocols to maintain accuracy and confidentiality.
- Digitize and restructure legacy paper records into a searchable relational database, improving data completeness and retrieval efficiency.
- Perform exploratory data analysis and build interactive Tableau dashboards, delivering usage-trend insights that inform administrative decisions.

Graduate Assistant, Probability & Statistics | *Clarkson University - Potsdam, NY Jan 2026 - May 2026*

- Tracked scheduling and mentorship-session data in a structured dataset, ensuring timely, accurate follow-up.
- Evaluated and tracked student performance data across assignments, maintaining a structured gradebook dataset of grades and outcomes.
- Partnered with faculty to deliver statistics instruction and academic support, reinforcing applied statistical concepts for undergraduate and graduate students.

Head Librarian | *Wise Owl Group of Schools- Zimbabwe May 2023 - Aug 2025*

- Implemented and administered the Koha Library Management System, migrating 1,000+ student and staff records into a centralized database under new data governance standards.
- Analyzed and visualized usage and inventory data to identify trends, directly informing budget, resource allocation, and technology investment decisions.
- Designed and enforced a data retention and secure-archiving policy, systematically purging outdated records to reduce data-security risk and maintain regulatory compliance.

Librarian | *Marondera High School - Zimbabwe Jul 2020 – Oct 2022*

- Queried and audited circulation, cataloging, and inventory records, identifying discrepancies and supporting data-informed inventory decisions.
- Analyzed circulation and usage records to identify borrowing trends, informing collection development decisions.

Executive Assistant | *Yellow Snake Mine - Zimbabwe Nov 2022 -Mar 2023*

- Sustained operational databases and structured records for executive reporting; coordinated scheduling and managed confidential correspondence and filing for senior leadership with discretion.

Information Officer | *Arathorn Engineering Services - Zimbabwe Dec 2018 - Mar 2020*

- Captured and organized structured employee, client, and receipt data; designed and introduced an electronic records-keeping system, migrating paper records into a secure digital database.

LEADERSHIP

- Head of Department, Wise Owl Group of Schools — led cross-departmental initiatives and information-sharing workflows.